

Database Technologies – Assignment 10

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, $\text{Perimeter} = 2 \times (L + W)$ and $\text{Area} = L \times W$.)

```
D6_96342_Jagdish@localhost> DELIMITER $$
D6_96342_Jagdish@localhost> CREATE PROCEDURE sp_rect(Wid INT, Len INT, OUT perim DOUBLE, OUT area DOUBLE)
-> BEGIN
->     SET perim = 2 * (Wid + Len);
->     SET area = Wid * Len;
-> END;
-> $$
```

Query OK, 0 rows affected (0.004 sec)

```
D6_96342_Jagdish@localhost>
D6_96342_Jagdish@localhost> DELIMITER ;
D6_96342_Jagdish@localhost> CALL sp_rect(10, 20, @per, @area);
Query OK, 0 rows affected (0.001 sec)
```

```
D6_96342_Jagdish@localhost> SELECT @per AS Perimeter, @area AS Area;
```

Perimeter	Area
60	200

1 row in set (0.000 sec)

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
D6_96342_Jagdish@localhost> CREATE TABLE temp(Number INT, NumSquare DOUBLE, NumCube DOUBLE);
Query OK, 0 rows affected (0.008 sec)
```

```
D6_96342_Jagdish@localhost> DELIMITER $$
D6_96342_Jagdish@localhost> CREATE PROCEDURE calc_num(num1 INT)
-> BEGIN
->   DECLARE num INT;
->   DECLARE numSquare DOUBLE;
->   DECLARE numCube DOUBLE;
->
->   SET num = num1;
->   SET numSquare = num1 * num1;
->   SET numCube = num1 * num1 * num1;
->
->   INSERT INTO temp VALUES(num, numSquare, numCube);
-> END;
-> $$
```

```
Query OK, 0 rows affected (0.003 sec)
```

```
D6_96342_Jagdish@localhost>
D6_96342_Jagdish@localhost> DELIMITER ;
D6_96342_Jagdish@localhost> CALL calc_num(9);
Query OK, 1 row affected (0.004 sec)
```

```
D6_96342_Jagdish@localhost> SELECT * FROM temp;
```

Number	NumSquare	NumCube
9	81	729

```
1 row in set (0.001 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```
D6_96342_Jagdish@localhost> DELIMITER $$
D6_96342_Jagdish@localhost> CREATE PROCEDURE convert_temp(temp DECIMAL, form VARCHAR(10))
-> BEGIN
->   DECLARE con_temp DOUBLE;
->   IF form = "F" THEN
->     SET con_temp = (temp - 32) * (5/9);
->     SELECT CONCAT(con_temp, ' Celcius') AS Converted_Temperature;
->   ELSEIF form = "C" THEN
->     SET con_temp = (9/5) * temp + 32;
->     SELECT CONCAT(con_temp, ' Fahrenheit') AS Converted_Temperature;
->   ELSE
->     SELECT "Please provide correct temperature form - F or C" AS message;
->   END IF;
-> END;
-> $$
Query OK, 0 rows affected (0.003 sec)

D6_96342_Jagdish@localhost>
D6_96342_Jagdish@localhost> DELIMITER ;
D6_96342_Jagdish@localhost> CALL convert_temp(37, 'C');
+-----+
| Converted_Temperature |
+-----+
| 98.6 Fahrenheit      |
+-----+
1 row in set (0.001 sec)

Query OK, 0 rows affected (0.001 sec)

D6_96342_Jagdish@localhost> CALL convert_temp(100, 'F');
+-----+
| Converted_Temperature |
+-----+
| 37.7808 Celcius      |
+-----+
1 row in set (0.001 sec)

Query OK, 0 rows affected (0.001 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
D6_96342_Jagdish@localhost> DELIMITER $$
D6_96342_Jagdish@localhost> CREATE PROCEDURE convert_inch_yard(inches INT)
-> BEGIN
->   DECLARE yards INT;
->   DECLARE feet INT;
->   DECLARE rem_inch INT;
->
->   SET yards = inches DIV 36;
->   SET feet = (inches MOD 36) DIV 12;
->   SET rem_inch = (inches MOD 36) MOD 12;
->
->   SELECT CONCAT(inches, ' inches equals ', yards, ' yards, ', feet, ' feet, and ', rem_inch, ' inches.') AS Inch_To_Yards;
-> END;
-> $$
Query OK, 0 rows affected (0.002 sec)

D6_96342_Jagdish@localhost>
D6_96342_Jagdish@localhost> DELIMITER ;
D6_96342_Jagdish@localhost> CALL convert_inch_yard(124);
+-----+
| Inch_To_Yards |
+-----+
| 124 inches equals 3 yards, 1 feet, and 4 inches. |
+-----+
1 row in set (0.001 sec)

Query OK, 0 rows affected (0.001 sec)

D6_96342_Jagdish@localhost> CALL convert_inch_yard(228);
+-----+
| Inch_To_Yards |
+-----+
| 228 inches equals 6 yards, 1 feet, and 0 inches. |
+-----+
1 row in set (0.000 sec)

Query OK, 0 rows affected (0.001 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
D6_96342_Jagdish@localhost> DELIMITER ;
D6_96342_Jagdish@localhost> DELIMITER $$
D6_96342_Jagdish@localhost> CREATE PROCEDURE prod_comp_100(num1 DOUBLE, num2 DOUBLE)
-> BEGIN
->     DECLARE prod DOUBLE;
->     SET prod = num1 * num2;
->
->     IF prod > 100 THEN
->         SELECT CONCAT("Product is greater than 100 -> ", prod) AS message;
->     ELSE
->         SELECT CONCAT("Product is less than 100 -> ", prod) AS message;
->     END IF;
-> END;
-> $$
Query OK, 0 rows affected (0.003 sec)

D6_96342_Jagdish@localhost>
D6_96342_Jagdish@localhost> DELIMITER ;
D6_96342_Jagdish@localhost> CALL prod_comp_100(12, 0.9);
+-----+
| message |
+-----+
| Product is less than 100 -> 10.8 |
+-----+
1 row in set (0.001 sec)

Query OK, 0 rows affected (0.001 sec)

D6_96342_Jagdish@localhost> CALL prod_comp_100(21, 4.9);
+-----+
| message |
+-----+
| Product is greater than 100 -> 102.9 |
+-----+
1 row in set (0.001 sec)

Query OK, 0 rows affected (0.001 sec)
```